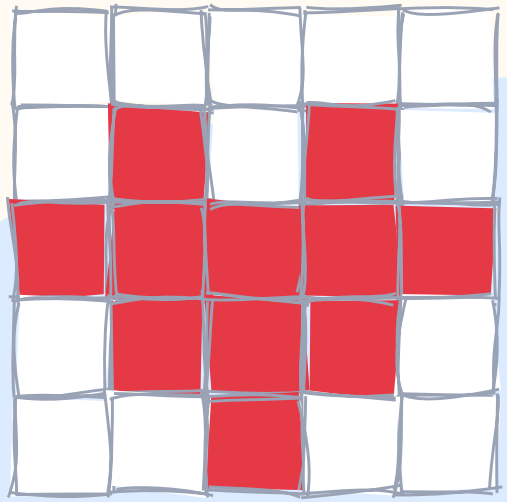
Little Explorers · Class 26

Pixel Messages

Send a picture using only numbers!

Pixels | Picture → numbers | Secret messages



0 0 0 0 0
0 1 0 1 0
1 1 1 1 1
0 1 1 1 0
0 0 1 0 0

Did you know?
When you send a photo on a phone, it travels as millions of numbers and turns back into a picture at the other end!

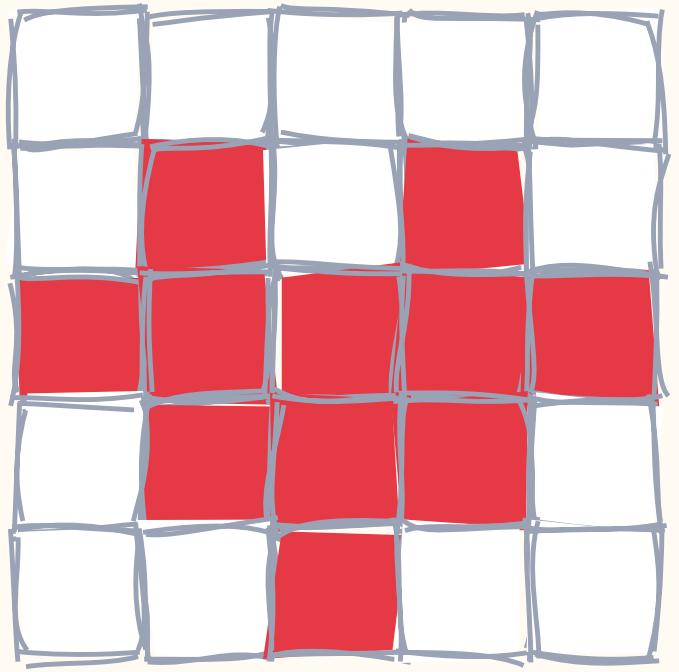
LEARN

Let's wonder...

How does a photo travel?

Whoosh!

Grandma sends a photo from far away, and it appears on your phone in a second. Pictures can't fly through wires! So what actually travels? **Numbers!**



Tiny pictures fly?

Magic?

Numbers travel!

Today: you turn pictures into **numbers** and send secret picture messages to a friend.

LEARN

Learning objectives

Today I can...

1		**Pixels:** say that pictures are made of tiny squares called **pixels**	**Proof:** I found pixels on a screen
2		**Picture to numbers:** turn a picture into **numbers** (1 = colour, 0 = empty)	**Proof:** I wrote my picture as numbers
3		**Send a message:** send my picture to a friend **using only numbers**	**Proof:** my friend drew my picture from my numbers

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Objective 1 · Pixels · video

Video time!



What Are Pixels?

Science Kids · about 2 minutes · youtu.be/1ptJaJT8jAM

Watch for...

1. What is a **pixel**?
2. How do lots of pixels make a picture?
3. Where can you find pixels?

Scan to watch

After the video: Look very close at a tablet screen. Can you see tiny dots?

Objective 1 · Pixels

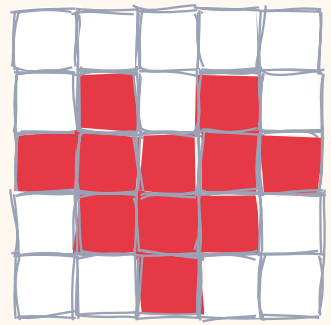
Pictures are made of pixels

The idea

Look very close at a screen: the picture is made of tiny **squares** called **pixels**.

Each pixel is just one colour. Lots of pixels together make a whole picture!

Example: a mosaic



Mosaics are pictures made of tiny coloured tiles. From far away you see a picture; up close you see squares. Screens work the same way!

Objective 2 · Picture to numbers

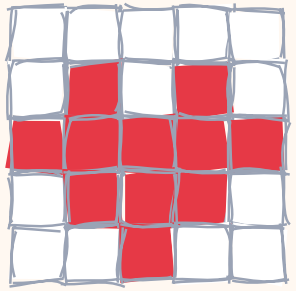
1 means colour, 0 means empty

The idea

Write a number for each square, row by row:
1 = colour it in. **0** = leave it white.

A heart becomes:
0 0 0 0 0 / 0 1 0 1 0 / 1 1 1 1 1 / 0 1 1 1 0 / 0 0 1 0 0

Example: knitting patterns



0 0 0 0 0
0 1 0 1 0
1 1 1 1 1
0 1 1 1 0
0 0 1 0 0

Knitting patterns use a grid of squares to show where each colour goes. Grandma reads the grid row by row!

LEARN

Objective 3 · Send a message

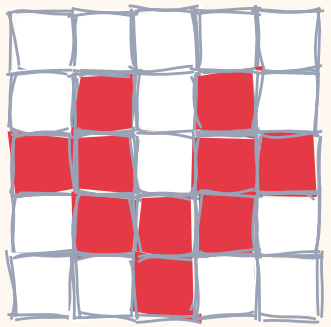
Mistakes in messages

The idea

If one number is wrong, one pixel is wrong! A 1 that becomes 0 makes a hole in the picture.

Computers **check** their messages to catch mistakes. We can check too: count the 1s in each row!

Example: whisper game



In the whisper game, a message gets mixed up as it passes along. Computers check messages carefully so pictures don't get mixed up.

Real world

Pictures travel as numbers



Photos
Sent as numbers on WhatsApp



TV
Every picture is pixels



Space
Mars rovers send photos as numbers



Games
Pixel art characters

? What pixel picture would you send to someone far away?

Space photos
Photos from Mars take minutes to reach Earth, travelling as numbers!

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Human pixels
2. Turn your picture into numbers
3. Send the message
4. Message test
5. Pixel challenges

MAKE

Class game

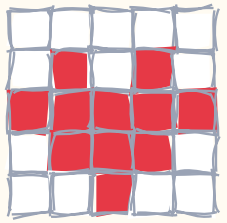
Human pixels

How to play

1. Make a 5 × 5 grid of children (or chairs).
2. Each child holds a card: **coloured side** or **white side**.
3. The teacher reads numbers: **row 1: 0 1 0 1 0**.
4. Show your side! What picture appears?

Then discuss

Could you make a picture with only 1s and 0s?



Big idea

Every picture can be made of tiny on/off squares.

Objective 2 · Build step 1 of 2

Turn your picture into numbers

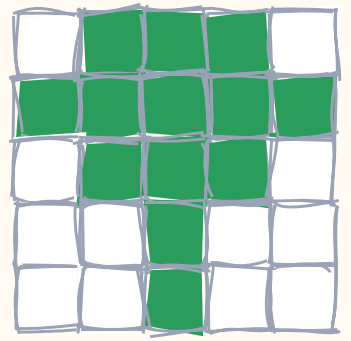
a Take the 5 × 5 grid sheet.

b Colour in squares to make a simple picture: a smile, a tree, a house.

c For each row, write **1** for coloured and **0** for white.

d Check: every row has 5 numbers!

e Fold the picture so nobody can see it.



0 1 1 1 0
1 1 1 1 1
0 1 1 1 0
0 0 1 0 0
0 0 1 0 0

Check: You have 5 rows of 5 numbers.

MAKE

Objective 3 · Build step 2 of 2

Send the message

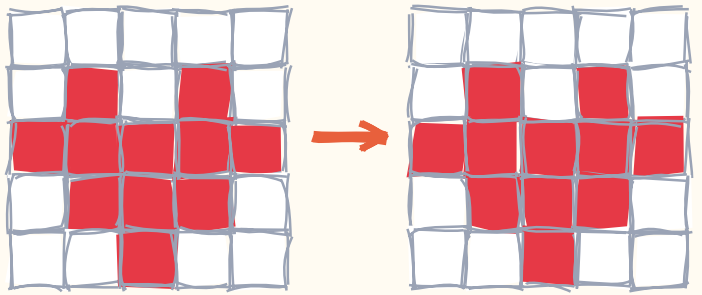
a Give ONLY your numbers to a partner. Hide your picture!

b Your partner colours an empty grid using your numbers.

c Open your picture. Do they match?

d Swap and try again.

e Try reading numbers out loud, like a radio message!



Check: Your partner's picture matches yours!

Test it!

Message test

Round	Matched?	Mistakes
Round 1		
Round 2 (read out loud)		
Round 3 (bigger picture)		

Talk about it

Was it easier to send numbers written or read out loud?

Count the 1s

Count the 1s in each row to check for mistakes.

MAKE

Level up!

Pixel challenges

Choose your challenge

- **Bigger grid:** try an 8 × 8 picture.
- **Colours:** use 2 = blue, 3 = green for more colours!
- **Class mural:** each team sends one part of a big picture.
- **Speed:** how fast can you send a message?

Think like an engineer

Why do phone photos need so many pixels?

Boss level

Send your initial as a pixel message!

Objective check

Can I really do it? Prove it.

01	What is a pixel?	○ ○ ○
02	Show your picture as 1s and 0s.	○ ○ ○
03	Did your partner draw your picture from your numbers?	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Pixel gallery: show both pictures side by side!

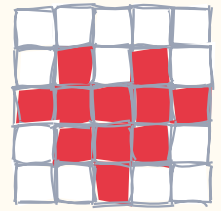
Show what you know

What did we learn?

- [x] Pictures are made of tiny pixels
- [x] 1 = colour, 0 = empty
- [x] Pictures travel as numbers
- [x] Check messages for mistakes

At home

Draw a pixel picture at home and send the numbers to a family member!



Exit ticket: what does 0 mean in our pixel message?

Next class: Pop the Bubbles with Your Hand!